Embedded Systems and IoT Overview

1. Introduction

Embedded systems are dedicated computer systems designed to perform specific tasks. The Internet of Things (IoT) connects physical devices to the internet for monitoring, control, and data exchange.

2. Types of Embedded Systems

• Standalone Embedded Systems
• Real-Time Embedded Systems (Hard and Soft)
• Networked Embedded Systems
• Mobile Embedded Systems

3. Types of IoT

• Consumer IoT
• Industrial IoT (IIoT)
• Commercial IoT
• Infrastructure IoT

4. IoT Deployment Models

• Cloud-based Deployment
• On-Premises Deployment
• Edge Computing Deployment
• Hybrid Deployment

5. Hardware Components

• Arduino Uno
• ESP8266 / ESP32
• Raspberry Pi
• Sensors (Temperature, Humidity, Ultrasonic, IR)
• HC-05 Bluetooth Module
• Wi-Fi Module
• Relay Module
• Motor Driver (L298N)
• DC Motors and Servo Motors
• Power Supply/Battery

6. Uses and Applications

• Smart Homes
• Smart Agriculture
• Healthcare Monitoring
• Industrial Automation
• Smart Cities
• Smart Vehicles
• Environmental Monitoring
• Security and Surveillance

7. Advantages

• Automation
• Remote Monitoring
• Improved Efficiency
• Cost Reduction
• Real-Time Data Collection

8. Conclusion

Embedded Systems and IoT are transforming industries by enabling intelligent automation, remote monitoring, and efficient resource management. They play a vital role in modern smart applications.